

DISCRETE MATHEMATICS AND STRUCTURES

PREDICATES AND QUANTIFIERS

Predicates

- ✧ A generalization of propositions - propositional functions or predicates: propositions which contain variables
- ✧ Predicates become propositions once every variable is bound- by
 - ★ assigning it a value from the Universe of Discourse U
 - ★ or quantifying it

Statements involving variables, such as:
 $x > 3$, $x = y + 3$, $x + y = z$
and *computer x is under attack by an intruder*
and *computer x is functioning properly*

Subjects and Predicates

- ✧ The statement " *x is greater than 3*" has 2 parts.
- ✧ The **1st part** = x is the subject of the statement
- ✧ The **2nd part** – the predicate = "*is greater than 3*" refers to a property that the subject of the statement can have.
The statement $P(x)$ – value of the **propositional function P** at x .

Examples:

- Let $U = \mathbb{Z}$, the integers = $\{\dots -2, -1, 0, 1, 2, \dots\}$
- ✧ $P(x)$: $x > 0$ is the predicate. It has no truth value until the variable x is bound.
- Examples of propositions where x is assigned a value:
- ✧ $P(-3)$ is false, $P(0)$ is false, $P(3)$ is true.
 - ✧ The collection of integers for which $P(x)$ is true are the positive integers.
 - ✧ $P(y) \vee \neg P(0)$ is not a proposition. The variable y has not been bound.

Quantifiers

Quantification expresses the extent to which a predicate is true over a range of elements. In English, the words *all*, *some*, *many*, *none*, and *few* are used in quantifications.

Two types of quantification: Universal and Existential

✧ **Predicate Calculus:** The area of logic that deals with predicates and quantifiers

✧ **Domain / Universe of discourse or**

Domain: A range for which values are considered to be covered by quantification

The Universal Quantifier

The universal quantification of $P(x)$ for a particular domain is the proposition that asserts that $P(x)$ is true for all values of x in this domain

- ✧ " *$P(x)$ for all values of x in the domain.*"
 - ✧ The notation $\forall xP(x)$ denotes the universal quantification of $P(x)$.
 - ✧ Here $\forall$ is called the universal quantifier.
 - ✧ We read $\forall xP(x)$ as "*for all x $P(x)$* " or "*for every x $P(x)$* ."
 - ✧ An element for which $P(x)$ is false is called a **counterexample** of $\forall xP(x)$.
- Example: $U = \{1, 2, 3\}$

$$\forall xP(x) \leftrightarrow P(1) \wedge P(2) \wedge P(3)$$

The Existential Quantifier

With existential quantification, we form a proposition that is true if and only if $P(x)$ is true for at least one value of x in the domain.

- ✧ "*There exists an element x in the domain such that $P(x)$.*"
 - ✧ We use the notation $\exists xP(x)$ for the existential quantification of $P(x)$.
 - ✧ Here $\exists$ is called the existential quantifier.
 - ✧ $\exists xP(x)$ is read as: "*There is an x such that $P(x)$.*" or "*There is at least one x such that $P(x)$.*" or "*For some x $P(x)$* "
- Example: $U = \{1, 2, 3\}$

$$\exists xP(x) \leftrightarrow P(1) \vee P(2) \vee P(3)$$

Unique Existential

$P(x)$ is true for one and only one x in the universe of discourse.

Notation: unique existential quantifier $\exists! x$
 $P(x)$

- ✧ "*There is a unique x such that $P(x)$,*"
"*There is one and only one x such that*

$P(x)$,” “One can find only one x such that $P(x)$.”

Remember: A predicate is not a proposition until all variables have been bound either by quantification or assignment of a value

Truth Table of Quantifiers

Statement	When True?	When False?
$\forall xP(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists xP(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

Negation	Equivalent Statements	When is Negation True?	When False?
$\neg \exists xP(x)$	$\forall x\neg P(x)$	For every x , $P(x)$ is false	There is an x for which $P(x)$ is true
$\neg \forall xP(x)$	$\exists x\neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x

Statement	When True?	When False?
$\forall x\forall yP(x, y)$ $\forall y\forall xP(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x\exists yP(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x for which $P(x, y)$ is false for every y .
$\exists x\forall yP(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x\exists yP(x, y)$ $\exists y\exists xP(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus.

✧ For example: $\forall xP(x) \vee Q(x)$ is the disjunction of $\forall xP(x)$ and $Q(x)$.

✧ In other words, it means $(\forall xP(x) \vee Q(x))$ rather than $\forall x(P(x) \vee Q(x))$.

✧ Quantify the proposition before quantifying the whole operation.

Translating from English into Logical Expressions

Express the statement “Every student in this class has studied calculus” using predicates and quantifiers.

Solution:

1. First, we rewrite the statement so that we can clearly identify the appropriate quantifiers to use. Doing so, we obtain: “For every student in this class, that student has studied calculus.”

2. Next, we introduce a variable x so that our statement becomes “For every student x in this class, x has studied calculus.”

3. Continuing, we introduce $C(x)$, which is the statement “ x has studied calculus.”

4. Consequently, if the domain for x consists of the students in the class, we can translate our statement as $\forall xC(x)$.

Negating Quantified Expressions

We will often want to consider the negation of a quantified expression.

✧ Example: “Every student in your class has taken a course in calculus.”

This statement is a universal quantification, namely, $\forall xP(x)$, where $P(x)$ is the statement “ x has taken a course in calculus” and the domain consists of the students in your class.

The negation of this statement is: “It is not the case that every student in your class has taken a course in calculus.”

This is equivalent to: “There is a student in your class who has not taken a course in calculus.”

And this is simply the existential quantification of the negation of the original propositional function, namely, $\exists x \neg P(x)$. This example illustrates the following logical equivalence: $\neg \forall x P(x) \equiv \exists x \neg P(x)$.

Example: Let $R(x)$ denote "x eats rice."
 ✱ Then the statement "All Asians eat rice" is represented by $\forall x R(x)$, where the domain consists of all Asians.
 ✱ The negation of this statement is $\neg \forall x R(x)$, which is equivalent to $\exists x \neg R(x)$.
 ✱ This negation can be expressed in several different ways, including "Some Asians do not eat rice" and "There is an Asian who does not eat rice."

Nested quantifiers

Multiple Quantifiers: read left to right...

Example: Let $U = \mathbb{R}$, the real numbers, $P(x, y)$: $xy = 0$

$\forall x \forall y P(x, y)$
 $\forall x \exists y P(x, y)$
 $\exists x \forall y P(x, y)$
 $\exists x \exists y P(x, y)$

✱ The only one that is false is the first one.

Example: Let $U = \{1, 2, 3\}$. Find an expression equivalent to $\forall x \exists y P(x, y)$ where the variables are bound by substitution instead:

✱ Expand from inside out or outside in.

Outside in:

$\exists y P(1, y) \wedge \exists y P(2, y) \wedge \exists y P(3, y) \leftrightarrow$
 $[P(1, 1) \vee P(1, 2) \vee P(1, 3)] \wedge [P(2, 1) \vee P(2, 2) \vee P(2, 3)] \wedge [P(3, 1) \vee P(3, 2) \vee P(3, 3)]$

Dangerous situations:

✱ *Commutativity of quantifiers*

$\forall x \forall y P(x, y) \leftrightarrow \forall y \forall x P(x, y)$? Yes

$\forall x \exists y P(x, y) \leftrightarrow \exists y \forall x P(x, y)$? No, different meaning

✱ *Distributivity of quantifiers over operators*

$\forall x [P(x) \wedge Q(x)] \leftrightarrow \forall x P(x) \wedge \forall x Q(x)$? Yes

$\forall x [P(x) \rightarrow Q(x)] \leftrightarrow [\forall x P(x) \rightarrow \forall x Q(x)]$? No

SET THEORY

✱ A **set** is a collection or group of objects or elements or members. (Cantor 1895)

✱ A set is said to contain its elements.

There must be an underlying universal set U , either specifically stated or understood.

Set Notation

Different ways of describing a set.

1. **Explicitly:** listing the elements of a set

✱ list the elements between braces: $\{1, 2, 3\}$ is the set containing "1" and "2" and "3."

✱ repetition is irrelevant.

✱ sets are unordered.

✱ use ellipses when the general pattern of the elements is obvious.

✱ $\{1, 2, 3, \dots\}$ is a way we denote an infinite set (in this case, the natural numbers)

✱ $\emptyset = \{\}$ is the empty set, or the set containing no elements.

2. **Implicitly:** by using set builder notations, stating the property or properties of the elements of the set.

✱ $S = \{m \mid 2 \leq m \leq 100, m \text{ is integer}\}$

✱ S is the set of all m such that m is between 2 and 100 and m is integer.

✱ specification by predicates: $S = \{x \mid P(x)\}$,
 ✱ S contains all the elements from U which make the predicate $P(x)$ true.

3. **Diagram:** by using a Venn Diagram

✱ $S = \{a, e, i, o, u\}$ which can also be represented as $S = \{v \mid v \text{ is a vowel in the English alphabet}\}$

Where:

✱ the rectangle represents U , the Universal Set which is the set containing all the letters in the English alphabet

✱ The circle represents the set S

✱ The dots inside the circle represent the elements in set S

Common Universal Sets

$\mathbb{R}$ = real numbers

$\mathbb{N}$ = natural numbers = $\{0, 1, 2, 3, \dots\}$, the counting numbers

$\mathbb{Z}$ = all integers $\{\dots, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$

$\mathbb{Z}^+$ = is the set of positive integers

Notation:

✱ x is a member of S or x is an element of S : $x \in S$.

✱ x is not an element of S : $x \notin S$.

Subsets

The set A is a subset of the set B , denoted $A \subseteq B$, iff $\forall x [x \in A \rightarrow x \in B]$

$A \subseteq B$ means " A is a subset of B ."

or, " B contains A "

or, "every element of A is also in B ."

or, $\forall x ((x \in A) \rightarrow (x \in B))$

$A \subseteq B$ means " A is a subset of B ."

$A \supseteq B$ means " A is a superset of B ."

$A = B$ if and only if A and B have exactly the same elements.

✳ iff, $A \subseteq B$ and $B \subseteq A$

✳ iff $A \supseteq B$ and $A \supseteq B$

✳ iff, $\forall x ((x \in A) \leftrightarrow (x \in B))$

So to show equality of sets A and B , show:

$A \subseteq B$ $B \subseteq A$

Definition: If $A \subseteq B$ but $A \neq B$ then we say A is a **proper subset** of B , denoted $A \subset B$.

$A \subseteq B$, and $A \neq B$.

$\forall x ((x \in A) \rightarrow (x \in B)) \wedge \neg \forall x ((x \in B) \rightarrow (x \in A))$

$\forall x ((x \in A) \rightarrow (x \in B)) \wedge \exists x (\neg(x \in B) \vee (x \in A))$

$\forall x ((x \in A) \rightarrow (x \in B)) \wedge \exists x ((x \in B) \wedge \neg(x \in A))$

$\forall x ((x \in A) \rightarrow (x \in B)) \wedge \exists x ((x \in B) \wedge (x \notin A))$

The void set, the null set, the empty set, denoted $\emptyset$, is the set with no members.

✳ Note: the assertion $x \in \emptyset$ is always false.

✳ Hence $\forall x [x \in \emptyset \rightarrow x \in B]$ is always true.

✳ Therefore, $\emptyset$ is a subset of every set.

✳ Note: A set B is always a subset of itself.

Set Cardinality

✳ The number of (distinct) elements in A , denoted $|A|$, is called the cardinality of A .

✳ If the cardinality is a natural number (in $\mathbb{N}$), then the set is called finite, else infinite.

Example: $A = \{a, b\}$

$$|\{a, b\}| = 2$$

If S is finite, then the cardinality of S , $|S|$, is the number of distinct elements in S .

Power Set

✳ Given a set S , the power set of S is the set of all subsets of S . The power set is denoted by $P(S)$.

✳ If a set has n elements, then the number of subsets in its power set is: $|P(S)| = 2^n$

This includes:

- ✳ the empty set ($\emptyset$)
- ✳ all possible combinations of elements
- ✳ the set itself

Examples:

∴ Assume an empty set $\emptyset$

∴ What is the power set of $\emptyset$? $P(\emptyset) = \{\emptyset\}$

∴ What is the cardinality of $P(\emptyset)$? $|P(\emptyset)| = 1$

Cartesian Product

∴ **Sets** are used to represent unordered collections.

∴ **Ordered-n tuples** are used to represent an ordered collection.

Definition: An ordered n -tuple $(x_1, x_2, \dots, x_N)$ is the ordered collection that has x_1 as its first element, x_2 as its second element, ..., and x_N as its N -th element, $N \geq 2$.

Definition: The Cartesian product of A with B , denoted $A \times B$, is the set of ordered pairs $\{ \langle a, b \rangle \mid a \in A \wedge b \in B \}$

Notation:

$$\times_{i=1}^n A_i = \{ \langle a_1, a_2, \dots, a_n \rangle \mid a_i \in A_i \}$$

$$A_1 \times A_2 \times \dots \times A_n = \{ \langle a_1, a_2, \dots, a_n \rangle : a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n \}$$

Note: The Cartesian product of anything with $\emptyset$ is $\emptyset$.

∴ Example: $A = \{a, b\}$, $B = \{1, 2, 3\}$

$$A \times B = \{ \langle a, 1 \rangle, \langle a, 2 \rangle, \langle a, 3 \rangle, \langle b, 1 \rangle, \langle b, 2 \rangle, \langle b, 3 \rangle \}$$

SET OPERATION

∴ Propositional calculus and set theory are both instances of an algebraic system called a **Boolean Algebra**.

∴ The operators in set theory are defined in terms of the corresponding operator in propositional calculus

∴ As always there must be a universe U . All sets are assumed to be subsets of U .

I. Let A and B be sets. The **union** of A and B, denoted by $A \cup B$, is the set that contains those elements that are either in A or in B, or in both.

∴ Alternate: $A \cup B = \{x \mid x \in A \vee x \in B\}$.

II. Let A and B sets. The **intersection** of A and B, denoted by $A \cap B$, is the set that contains those elements that are in both A and B.

∴ Alternate: $A \cap B = \{x \mid x \in A \wedge x \in B\}$

III. Two sets are called **disjoint** if their intersection is empty.

∴ Alternate: A and B are disjoint if and only if $A \cap B = \emptyset$

Cardinality of the set union

$$|A \cup B| = |A| + |B| - |A \cap B|$$

IV. Let A and B be sets. The **difference** of A and B, denoted by $A - B$, is the set $A \cap \bar{B}$ containing those elements that are in A but not in B. The difference of A and B is also called the **complement of B with respect to A**.

∴ Alternate: $A - B = \{x \mid x \in A \wedge x \notin B\}$

V. The **symmetric difference**, $A \oplus B$, is:

$$A \oplus B = \{x : (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$$

$$= (A - B) \cup (B - A)$$

∴ Like "exclusive or"

VI. The **complement** of a set A is: $A = \{x : x \notin A\}$

∴ *Alternative notation is A^c ,*

∴ *The (absolute) complement of A is $U - A$,* where U is the universal set.

∴ $\bar{\emptyset} = U$ and $\bar{U} = \emptyset$

Examples:

$$U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 3, 4, 5\}, B = \{4, 5, 6, 7, 8\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A \cap B = \{4, 5\}$$

$$\bar{A} = \{0, 6, 7, 8, 9, 10\}$$

$$\bar{B} = \{0, 1, 2, 3, 9, 10\}$$

$$A - B = \{1, 2, 3\}$$

$$B - A = \{6, 7, 8\}$$

$$A \oplus B = \{1, 2, 3, 6, 7, 8\}$$

Set Identities

Methods for Proving Set Identities

∴ Show that each is a subset of the other

∴ Using set builder notation and logical equivalences

∴ Using Membership tables

Identity	Name
$A \cup \emptyset = A$ $A \cap U = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\bar{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap C$ $A \cap (B \cup C) = (A \cap B) \cup C$	Associative laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$\overline{A \cup B} = \bar{A} \cap \bar{B}$ $\overline{A \cap B} = \bar{A} \cup \bar{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	Complement laws

Example:

∴ Using set builder notation and logical equivalences

∴ Use set builder notation and logical equivalences to establish the second De

Morgan law $\overline{A \cap B} = \bar{A} \cup \bar{B}$

Solution: We can prove this identity with the following steps.

$\overline{A \cap B}$	$= \{x \mid x \notin A \cap B\}$	by definition of complement
	$= \{x \mid \neg(x \in (A \cap B))\}$	by definition of does not belong symbol
	$= \{x \mid \neg(x \in A \wedge x \in B)\}$	by definition of intersection
	$= \{x \mid \neg(x \in A) \vee \neg(x \in B)\}$	by the first De Morgan law for logical eq
	$= \{x \mid x \notin A \vee x \notin B\}$	by definition of does not belong symbol
	$= \{x \mid x \in \overline{A} \vee x \in \overline{B}\}$	by definition of complement
	$= \{x \mid x \in \overline{A} \cup \overline{B}\}$	by definition of union
	$= \overline{A \cap B}$	by meaning of set builder notation

